

Gyuwon Na

uoyuvvou@gmail.com | +82 10-6409-8658 | [linkedin.com/in/uoyuvvou](https://www.linkedin.com/in/uoyuvvou) | github.com/Gyuwon-Na | gyuwon-na.github.io

EDUCATION

DigiPen Institute of Technology | Redmond, WA

Aug 2026 – Expected May 2028

B.S. Computer Science in Real-Time Interactive Simulation | GPA: 4.0 / 4.0

Keimyung University | Daegu, South Korea

Mar 2023 - Jun 2026

KMU-DigiPen Dual-Degree Program | GPA: 4.38 / 4.5

Coursework: Computer Graphics, Data Structures, Computer Networks, Digital Signal Processing, Circuit Analysis

TECHNICAL SKILLS

Languages: C++, Python, C, GLSL

Tools & Platforms: Linux, CMake, Git/GitHub, ROS 2 / ROS 1, Docker, MATLAB

Libraries & Engines: OpenGL, WebGL, OpenCV, PyTorch, CUDA, Unity

RESEARCH EXPERIENCE

ISIP Lab | Undergraduate Researcher | Daegu, South Korea

Aug 2025 - Present

- Designed a **1.43M-parameter, five-stage geometric alignment framework** using Clifford-inspired scalar, vector, and rotor-related features with transformer, residual, and energy-based refinement for severe in-plane rotations.
- Reduced mean rotation error on the 100 hardest cases from **40.49° to 17.01° (58%)**; built COCO, HPatches, and MegaDepth evaluations with baseline, ablation, and failure-case analysis.
- First author of an **Image and Vision Computing** manuscript under review
- First author of a **KMMS Fall Conference poster** on adaptive binarization-based lane detection.

PROJECTS

SEA:ME Hackathon - Vision-Only Driving System | Lead Autonomy Engineer

Jul 2026

- Led end-to-end integration of a **single-camera ROS 2 autonomy stack** spanning lane perception, NCNN detection, mission state machines, Pure Pursuit control, debugging tools, and embedded actuator interfaces.
- Increased end-to-end perception from **6 FPS to a stable 20 FPS (3.3x)** through shared-memory process isolation, mission-specific ROIs, fresh-frame queues, and ROS 2 execution tuning.
- Won the **Silver Award**, placing **4th overall** with a 56-second run - only **0.03 seconds** behind 3rd place.

KAIST Mobility Challenge - V2X Cooperative Driving | Systems Engineer

Oct 2025 - Feb 2026

- Built V2X cooperative decision and control logic for **four autonomous vehicles** negotiating roundabouts and four-way intersections, including stop-or-go collision avoidance from shared pose and velocity state.
- Implemented MPC-inspired Pure Pursuit tracking, sim-to-real braking compensation, and YAML-configured vehicle tuning; earned **3rd place** and the **Excellence Award**.

MANZO - Custom C++ / OpenGL Game Engine | Technical Lead

Jun 2024 - Jun 2025

- Led a **year-long engine and game project**, implementing vector-font rendering, underwater visibility shaders, reusable UI systems, positional audio, and rendering diagnostics with RenderDoc.
- Built autonomous fish behaviors with pathfinding, terrain avoidance, flocking, and **quadtree-based spatial partitioning** to reduce per-frame spatial queries.

LEGO Finder - Prompt-Guided Visual Search | Computer Vision Project

Apr 2026 - Jun 2026

- Implemented a real-time, **prompt-guided LEGO finder** that narrows search with ROI masking, queries an offline **HNSW** index, verifies matches with RANSAC, and highlights targets in AR.

LEADERSHIP EXPERIENCE

Keimyung University | Class Representative & Student Council Member

Sep 2023 - Jun 2025

- Represented students as the primary liaison with faculty, consolidating academic concerns and communicating decisions across the dual-degree program.
- Coordinated academic events, student activities, and departmental programs while aligning schedules and responsibilities across students, faculty, and organizers.